근거를 제시하지 않음	*(0,5)*
$L = \frac{1}{\omega^2 C}$	0,7
$L = 10^{-2}$ Hn	0,3
$R = r$	0,5
$P_{max} = \frac{U^2}{8r}$	0,3
$P_{max} = 14$ mW	0,2
합계	**3,0**

문제 2. 전기장 속의 도체 (10,0점)

도체 구와 점전하

2.1 q

$$\varphi_1 = \frac{q}{4\pi\varepsilon_0\sqrt{(l-x)^2+y^2}}, \quad (1)$$

Q

$$\varphi_2 = \frac{Q}{4\pi\varepsilon_0\sqrt{(x-a)^2+y^2}}. \quad (2)$$

(1) (2)

$$\varphi = \varphi_1 + \varphi_2 = \frac{q}{4\pi\varepsilon_0\sqrt{(l-x)^2+y^2}} + \frac{Q}{4\pi\varepsilon_0\sqrt{(x-a)^2+y^2}}. \quad (3)$$

2.2

$$x^2 + y^2 = R^2. \quad (4)$$

y (4) (3)

$$\varphi = \frac{q}{4\pi\varepsilon_0\sqrt{l^2-2lx+R^2}} + \frac{Q}{4\pi\varepsilon_0\sqrt{a^2-2ax+R^2}}. \quad (5)$$

2.3 . ,

$$\varphi = 0. \quad (6)$$

(5)

$$\frac{Q}{q} = -\frac{\sqrt{a^2-2ax+R^2}}{\sqrt{l^2-2lx+R^2}} = \beta = const < 0. \quad (7)$$

(7)

$$2x(l\beta^2 - a) + a^2 + R^2 - \beta^2(l^2 + R^2) = 0. \quad (8)$$

(8) $x \in (-R, R)$.

x . ,

$$l\beta^2 - a = 0, \quad (9)$$

$$a^2 + R^2 - \beta^2(l^2 + R^2) = 0. \quad (10)$$

(9) (10)

$$a = l, \quad \beta = -1, \quad (11)$$

$$a = \frac{R^2}{l}, \quad \beta = -\frac{R}{l}. \quad (12)$$

(12) ,

$$Q = -q\frac{R}{l}, \quad (13)$$

$$a = \frac{R^2}{l}. \quad (14)$$

2.4

$$F = \frac{q^2 Rl}{4\pi\varepsilon_0(l^2-R^2)^2}, \quad (15)$$

$$A = \int_l^\infty F dl = \frac{q^2 R}{8\pi\varepsilon_0(l^2-R^2)}. \quad (16)$$

2.5 .

, .

W_q q , W_Q